PAPER_286: M16 Eagle Nebula UQFF — Nebular Friedmann Redshift Parameter κ_neb

First UQFF Nebular Module with z > 0: H(z=0.0015) = 70.047 km/s/Mpc

Classification: UQFF 2.0 Gravitational Physics — Nebular Cosmological Coupling
System: M16 Eagle Nebula (IC 4703), ~5700 ly distance, z = 0.0015
Session: 80 | **Module:** M16_UQFF_MODULE.cpp (22nd C++ UQFF module)
Author: Daniel T. Murphy | **Date:** March 2026

Abstract

M16 (Eagle Nebula, IC 4703) is located at ~5700 light-years distance, corresponding to cosmological redshift z = 0.0015. While this redshift is sub-Hubble (far smaller than extragalactic objects), it is **non-zero**, making M16 the **first UQFF nebular module to carry a cosmological redshift parameter**. This paper defines the **UQFF Nebular Redshift Parameter** κ_neb = [H(z=0.0015) − H(z=0)] / H(z=0) = **6.71×10^{-4}**, quantifying the fractional departure of the Friedmann Hubble rate at M16's redshift from the present-epoch value. The expansion coupling term g_exp = g_base × H(z=0.0015) × t contributes g_exp ≈ 5.21×10^{-16} m/s² at t = 5 Myr — tiny but formally catalogued for the first time in UQFF nebular physics.

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2. Physical Motivation

All prior UQFF nebular modules (Session 55 M16 in CP3) set z = 0, ignoring the cosmological correction from the Eagle Nebula's finite distance. However, the full Friedmann H(z) formalism correctly accounts for the **look-back time effect**: at z = 0.0015, the Hubble flow was marginally faster than today. The UQFF Nebular Redshift Parameter κ_neb quantifies this correction for the M16 Hubble expansion coupling term.

Prior UQFF modules with z > 0 were all **extragalactic** (galaxies, galaxy clusters). This paper extends the z > 0 treatment to the **sub-kiloparsec, sub-Hubble-flow nebular regime** for the first time.

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3. Mathematical Formulation

3.1 Friedmann H(z)

The Flat ΛCDM Friedmann equation for the Hubble parameter at redshift z:

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

Parameters: | Parameter | Value | |———-|——-| | H_0 | 70.0 km/s/Mpc | | Ω_m | 0.3 | | Ω_Λ | 0.7 | | z (M16) | 0.0015 |

H(z = 0):

$$H(0) = 70.0 \times \sqrt{0.3 \times 1^3 + 0.7} = 70.0 \times \sqrt{1.0} = 70.000 \text{ km/s/Mpc}$$

H(z = 0.0015):

$$(1 + 0.0015)^3 = 1.00451$$

$$0.3 \times 1.00451 + 0.7 = 1.00135$$

$$H(0.0015) = 70.0 \times \sqrt{1.00135} = 70.0 \times 1.000675 = \mathbf{70.047} \text{ km/s/Mpc}$$

3.2 UQFF Nebular Redshift Parameter κ_neb

$$\boxed{\kappa_{neb} = \frac{H(z = 0.0015) - H(z = 0)}{H(z = 0)} = \frac{70.047 - 70.000}{70.000} = \frac{0.047}{70.000} = 6.71 \times 10^{-4}}$$

This fractional correction is **small but physically meaningful** — it encodes the sub-Hubble cosmological velocity gradient at M16's position in the Milky Way's extended gravitational domain.

3.3 Friedmann Expansion Gravity Coupling

The Hubble expansion coupling on the nebula's self-gravity:

$$g_{exp}(t) = g_{base} \times H(z = 0.0015) \times t$$

Converting H(z=0.0015) to SI (s^{-1}):

$$H_{SI} = 70.047 \text{ km/s/Mpc} \times \frac{10^3 \text{ m/km}}{3.086 \times 10^{22} \text{ m/Mpc}} = 2.270 \times 10^{-18} \text{ s}^{-1}$$

At t = 5 Myr = 1.578×10^{14} s:

$$g_{exp}(5 \text{ Myr}) = 1.454 \times 10^{-12} \times 2.270 \times 10^{-18} \times 1.578 \times 10^{14}$$

$$= 1.454 \times 10^{-12} \times 3.582 \times 10^{-4} = \mathbf{5.21 \times 10^{-16}} \text{ m/s}^{\mathbf{2}}$$

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4. Comparison Across UQFF Nebular / Stellar Modules

Module	z	κ_type	H(z)
SATURN (Session 78)	0	—	70.000 km/s/Mpc
Prior M16 CP3 (Session 55)	0	—	70.000 km/s/Mpc
M16 THIS MODULE (PAPER_286)	**0.0015**	**κ_neb = $6.71{\times}10^{-4}$**	**70.047 km/s/Mpc**
ANDROMEDA (Session 76)	~0	κ_recession	70.000 km/s/Mpc
Extragalactic modules	> 0.001	κ_recession	> 70.0 km/s/Mpc

M16 is the **first UQFF module** to formally assign a z > 0 value to a **nebular (sub-galactic)** object, introducing κ_neb as a distinct parameter class from the galactic/extragalactic κ_recession used in prior modules.

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5. Physical Significance

Why z = 0.0015 is Not Simply z = 0

1. **Formal completeness:** The UQFF framework demands that all physical parameters be carried with their correct values. Setting z = 0 for M16 discards a real (if small) cosmological correction.
2. **Template for nearby nebulae:** The sub-Hubble-flow regime (z < 0.01) is occupied by many galactic star-forming regions — Orion (z ≈ 0.0014), Carina (z ≈ 0.0026), Rho Ophiuchi (z ≈ 0.0004). κ_neb provides a universal scaling parameter for this entire class.
3. **Detectability threshold:** κ_neb = 6.71 × 10^{-4} is above the precision floor of modern distance measurements (Gaia DR3: ≤ 0.01% parallax errors for nearby nebulae), meaning the H(z) correction is formally measurable.

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6. UQFF Module g_exp Context

In the full M16 UQFF 2.0 equation:

$$g_{total}(r,t) = \left[g_{dyn}(t) + U_{g,sum}(26) + \Lambda + Q + L + F + g_{exp}\right] \times \mathrm{corr}_{SC}$$

g_exp uses H(z=0.0015) rather than H(z=0). The fractional difference in g_exp due to κ_neb:

$$\frac{\Delta g_{exp}}{g_{exp}} = \kappa_{neb} = 6.71 \times 10^{-4}$$

This is a 0.067% correction to g_exp, which is itself a tiny term — but it is catalogued precisely as the UQFF Nebular Friedmann correction.

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7. Wolfram KB Term

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M16UQFF:kappa_neb=[H[z=0.0015]-H[z=0]]/H[z=0]=6.71e-4; H[z=0.0015]=70.047 km/s/Mpc [PAPE
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8. Cross-References

- **PAPER_284:** Dual Mass Co-Action Product (Φ_dm)
- **PAPER_285:** Erosion Saturation Half-Time (t_half, ΔgMax)
- **M16_UQFF_MODULE.cpp:** Full UQFF 2.0 C++ implementation (22nd module, first nebular z>0)
- **CondensedPhysics3.py:** `M16NebularFriedmannRedshiftCalculator`
